Online Quantum Multi-Path Selection Under Stochastic Blocking

***Abstract*—Quantum path selection directly affects the throughput, resource efficiency, and reliability of entanglement distribution in quantum networks. Previous studies often assume that path parameters, such as ebit rates, costs, and fidelities, are known and that paths remain available throughout communication. In practice, however, these parameters are typically unknown and must be learned online. Moreover, a quantum path can become blocked for a random duration after use, leading to suboptimal or infeasible selections. In this paper, we consider an online quantum multi-path selection problem that accounts for stochastic path blocking and maximizes the total weighted ebit rate subject to a cost budget and a fidelity threshold. We propose a bandit-learning algorithm, Online Quantum Multi-Path Selection (OQMPS), and prove that it achieves sublinear regret and sublinear constraint violation. Simulations on a 50-path quantum network show that OQMPS achieves near-zero long-term violation and reward performance competitive with an ideal benchmark.**

***Index Terms*—Quantum networks, multi-armed bandit, multi-path selection, stochastic blocking**

I. Introduction

In quantum networks, path selection between source-destination (SD) pairs plays a central role in determining the ebit rate of entanglement distribution, the efficient use of scarce quantum resources, and the fidelity of delivered entanglement. It therefore supports applications such as distributed quantum computing [1] and quantum sensing [2]. In practice, path selection is constrained by probabilistic entanglement generation, costly quantum memories, and the requirement that path fidelity remain above a specified threshold.

Existing works on quantum network path selection, however, exhibit several limitations. First, many approaches assume that the statistics of ebit rates, costs, and fidelities are known a priori and focus on offline optimization under full information [3], [4], whereas in practice these quantities are unknown and depend on time-varying channel conditions, resource contention, and hardware noise. Second, while learning-based approaches address this unknown-statistics challenge [5]–[7], they typically do not enforce budget and fidelity constraints. Third, existing works often neglect the intrinsic *stochastic blocking* of quantum paths: after a path is used for entanglement distribution, its quantum resources are consumed and require a non-negligible time to be re-established, during which the path is unavailable for future requests. This recovery time is random and governed by the success probabilities of entanglement generation and swapping, which determine the number of attempts required before the path is re-established [8], [9].

To address the limitations above, we consider an online path selection problem that accounts for stochastic path blocking. The objective is to learn the path parameters and select multiple paths for SD pairs to maximize the total weighted rate, subject to a cost budget and a fidelity threshold. The cost budget reflects limited quantum resources, such as entanglement consumption or quantum-memory occupation; exceeding it risks resource exhaustion or network congestion. The fidelity threshold is required because a quantum state with fidelity below the threshold becomes unusable for applications such as entanglement distillation [10] or quantum key distribution [11].

To solve this problem, we propose an online learning algorithm, Online Quantum Multi-Path Selection (OQMPS). The algorithm first observes the set of available (unblocked) paths. For each path, it maintains empirical averages of the ebit rate, cost, and fidelity and constructs optimistic estimates for the rate and fidelity together with a pessimistic estimate for the cost. The algorithm then invokes an approximation oracle over the available paths, which returns a subset of paths whose estimated cost and fidelity meet the constraints while approximately maximizing the estimated ebit rate. After the paths have been used, the algorithm updates the corresponding statistics with the observed rate, cost, and fidelity. Each selected path then becomes blocked for a random duration while its resources are re-established, during which it remains unavailable. We prove that OQMPS achieves sublinear regret relative to an optimal dynamic policy and sublinear total long-term violation that aggregates both cost overruns and fidelity shortfalls. Simulations show that OQMPS achieves reward performance competitive with an ideal benchmark and near-zero total long-term violation. Compared with the baselines, OQMPS attains the highest reward among constraint-aware algorithms while keeping violation growth sublinear. More simulations under varying cost budgets, fidelity thresholds, and delays further confirm its robustness.

The main contributions of this paper are: (i) We formulate an online quantum multi-path selection problem that maximizes the weighted rate subject to a per-slot cost budget and a fidelity threshold, and we propose OQMPS, an online learning algorithm that accommodates stochastic blocking and selects feasible subsets for multiple SD pairs. (ii) We prove that OQMPS achieves sublinear regret and sublinear total long-term violation, where violation aggregates both cost overruns and fidelity shortfalls. (iii) Simulations on a 50-path quantum network show that OQMPS achieves near-zero long-term violation and reward performance competitive with an ideal benchmark, and

is robust under varying parameters.

The remainder of this paper is organized as follows. Sec. II reviews related work. Sec. III formulates the problem. Sec. IV presents the online algorithm and its theoretical analysis. Sec. V reports simulation results. Sec. VI concludes the paper and discusses future work.

II. Related Work

Quantum network path selection has been studied from several perspectives. A major line of work focuses on ebit-rate or throughput maximization. Early approaches assume known statistics and adapt classical shortest-path algorithms to quantum repeater networks [3]. More recent works address this objective under fidelity constraints, including fidelity-based path selection for sum-rate maximization under a fidelity threshold [12], redundant entanglement provisioning and selection for throughput maximization [13], and online path selection under fidelity thresholds via mean-variance rewards [6]. Another line of work studies resource costs and budget constraints. Adaptive user-centric path selection maximizes entanglement success rate under a long-term budget via per-slot Lyapunov optimization [14], while bounded-cost successive elimination addresses cost-aware link selection [15]. Finally, temporal path unavailability has been characterized by blocking probability in mixed-efficiency repeater networks [16], and protection-path-based resource allocation has been studied to ensure service availability in QKD networks [17]. These works, however, typically assume either known statistics or single-shot offline optimization.

A commonly used framework for path selection is bandit learning. Introduced by Chen et al. [18], the combinatorial multi-armed bandit (CMAB) framework provides a principled foundation for sequential decision-making under combinatorial constraints and has been extended to semi-bandit feedback [19], general reward functions [20], and approximation oracles for NP-hard offline problems [21]. Classical CMAB, however, focuses on per-round reward maximization without enforcing constraints that accumulate over time. Liu et al. [22] address this limitation by introducing combinatorial bandits with linear long-term constraints and establishing logarithmic regret and zero expected violation. Aggarwal et al. [23] propose a bi-criteria framework that converts offline approximation algorithms into online CMAB algorithms with sublinear regret and cumulative constraint violation guarantees. Unlike prior bandit frameworks, we account for stochastic blocking of arms caused by resource recovery.

Other related but distinct models include sleeping bandits [24], in which arms may be unavailable in certain rounds owing to exogenous factors, and mortal bandits [25], in which arms have finite lifetimes. The blocking-bandit model is closest to our work: arms become unavailable for a random duration after each pull. Basu et al. [26] introduced deterministic blocking; Atsidakou et al. [27] extended it to combinatorial actions with stochastic delays, proving greedy approximation ratios and regret bounds. Although blocking bandits consider recovery delays, they neglect the long-term cost of arms and application-specific constraints.

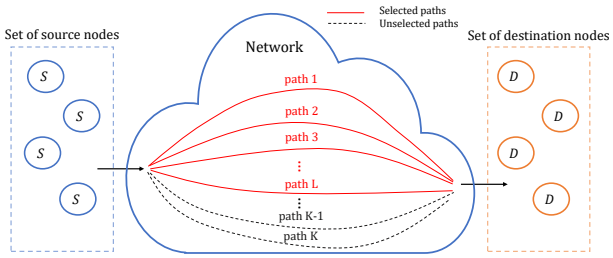


Fig. 1. Quantum network model for online path selection.

III. Problem Formulation

As shown in Fig. 1, we consider a quantum network with sets of source and destination nodes. The network has K pre-computed end-to-end paths connecting source and destination nodes. Let $[K] = \{1, 2, \ldots, K\}$ denote the set of all paths. We consider a discrete-time model in which time is partitioned into slots indexed by $t = 1, 2, \ldots, T$, where T is the time horizon. Each time slot corresponds to one round of path selection and quantum operation. Let $[T] = \{1, 2, \ldots, T\}$ denote the set of all time slots.

At each time slot t, the network selects L of the K paths ($L < K$) to serve L concurrent connection requests from the K SD pairs. Each selected path is assigned to exactly one request. The set of selected paths at time slot t is denoted by $\mathcal{P}_t \subset \{1, 2, \ldots, K\}$ with $|\mathcal{P}_t| = L$. Each path i has a positive priority weight w_i, which may represent the importance of the SD pair it serves or the service level of the traffic it carries; paths with higher weights have higher priority. For each selected path $i \in \mathcal{P}_t$, the observed ebit rate $r_i(t) \in [0, 1]$, usage cost $\mu_i(t) \in [0, 1]$, and end-to-end fidelity $F_i(t) \in [0, 1]$ are modeled as random variables drawn from fixed but unknown distributions. Their expectations are $r_i = \mathbb{E}[r_i(t)]$, $\mu_i = \mathbb{E}[\mu_i(t)]$, and $F_i = \mathbb{E}[F_i(t)]$. We assume that the rate, cost, and fidelity of a path are mutually independent. For convenience, we define the vectors of these quantities over all paths as $\boldsymbol{r}(t) = (r_1(t), \ldots, r_K(t))$, $\boldsymbol{\mu}(t) = (\mu_1(t), \ldots, \mu_K(t))$, and $\boldsymbol{F}(t) = (F_1(t), \ldots, F_K(t))$.

Whenever path i is selected at time slot t, it becomes blocked (unavailable) for $D_i(t) \in \mathbb{N}_{\geq 0}$ subsequent slots, that is, it cannot be selected again within the interval $[t, \ldots, t + D_i(t))$. We model the blocking time $D_i(t)$ as a random variable drawn from a fixed but unknown path-dependent distribution with mean $d_i = \mathbb{E}[D_i(t)]$ and support in $[0, D_{\max}]$. A path with $D_i(t) = 0$ experiences no blocking, and $D_{\max}$ denotes the maximum delay.

We next formulate the online quantum multi-path selection problem.

Objective. The total weighted ebit rate at time slot t is $\sum_{i \in \mathcal{P}_t} w_i r_i(t)$, where w_i is the priority weight of path i.

Our objective is to select paths that maximize the long-term average total weighted rate over T time slots.

$$\max_{\mathcal{P}_t, t\in[T]} \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t} w_i r_i(t). \tag{1}$$

Availability constraint. Let $\mathcal{A}_t = \{i \in [K] \mid \forall s < t, \mathbb{I}(i \in \mathcal{P}(s)) \cdot \mathbb{I}(D_i(s) \geq t - s) = 0\}$ denote the set of available (unblocked) paths at the beginning of time slot t, where $\mathbb{I}(\cdot)$ is the indicator function that equals 1 when its argument is true and 0 otherwise. Selected paths must be available (unblocked). That is,

$$\mathcal{P}_t \subseteq \mathcal{A}_t,\ \forall t \in [T]. \tag{2}$$

Budget constraint. The total cost incurred by the selected paths must not exceed the given budget B at every time slot:

$$\sum_{i\in\mathcal{P}_t} \mu_i(t) \leq B,\ \forall t \in [T]. \tag{3}$$

Fidelity constraint. To ensure the quality of quantum communication, each selected path must satisfy the fidelity threshold at every time slot:

$$F_i(t) \geq F_{\min},\ \forall i \in \mathcal{P}_t, t \in [T], \tag{4}$$

where the threshold $F_{\min}$ denotes the minimum fidelity requirement and satisfies $F_{\min} \geq 0.5$. A fidelity below 0.5 cannot certify the presence of entanglement [28].

To establish a theoretical performance benchmark, we consider an optimal dynamic policy Π^* that knows the distributions of $r_i(t)$, $\mu_i(t)$, $F_i(t)$ and $D_i(t)$. This policy is dynamic because its decisions must adapt to the stochastically evolving set of available paths, which changes over time owing to prior path selections and random blocking durations. Let $\mathcal{P}_t^*$ denote the set of paths selected by policy Π^* at time slot t. Let z_i denote the expected selection rate of path i under the optimal dynamic policy Π^*, given by

$$z_i = \mathbb{E}\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{I}(i \in \mathcal{P}_t^*)\right]. \tag{5}$$

The optimal expected cumulative reward achieved by Π^* is

$$\mathrm{Rew}^*(T) = T\sum_{i=1}^{K} w_i r_i z_i. \tag{6}$$

At each time slot t, an online algorithm selects a set $\mathcal{P}_t$ of L paths based solely on the partial observations from its previous selections. The algorithm does not know the underlying distributions of the path parameters and observes only the actual rate, cost, and fidelity of the selected paths; no observations are received for the unselected paths. Thus, the algorithm must learn these parameters while selecting paths.

In the full-information setting, where the true distributions of rate, cost, fidelity, and blocking delays are known and only the stochastic realizations and the evolving availability set remain uncertain, the quantum multi-path selection defined by (1)–(4) is a combinatorial optimization problem. It is well established that this problem is NP-hard, and finding an exact solution is intractable. Hence, rather than assume an exact solver, we introduce the following (α, β)-approximation oracle, which returns a feasible near-optimal solution with high probability:

Definition 1 ((α, β)-approximation oracle)**.** At time slot t, given a set of available paths $\mathcal{A}_t$ with $|\mathcal{A}_t| > L$, a reward vector $\boldsymbol{r}$, a cost vector $\boldsymbol{\mu}$, and a fidelity vector $\boldsymbol{F}$, denote the superset $\mathfrak{P}(\mathcal{A}_t) = \{\mathcal{P} \subseteq \mathcal{A}_t : |\mathcal{P}| = L, \sum_{i\in\mathcal{P}} \mu_i \leq B, F_i \geq F_{\min}, \forall i \in \mathcal{P}\}$. $f^{\alpha,\beta}(\mathcal{A}_t, \boldsymbol{r}, \boldsymbol{\mu}, \boldsymbol{F})$ is an (α, β)-approximation oracle that yields a set of paths $\mathcal{Q} \in \mathfrak{P}(\mathcal{A}_t)$ such that

$$\Pr\left[\sum_{i\in\mathcal{Q}} w_i r_i \geq \alpha \cdot \max_{\mathcal{P}\in\mathfrak{P}(\mathcal{A}_t)} \sum_{i\in\mathcal{P}} w_i r_i\right] \geq \beta. \tag{7}$$

A representative instance of such an oracle is a fully polynomial-time approximation scheme (FPTAS) [29], whose parameters satisfy $\alpha = 1 - \epsilon$ and $\beta = 1$, where $\epsilon > 0$ controls the trade-off between approximation accuracy and computational cost. We consider a feasible, approximately optimal algorithm, $\widetilde{\mathrm{OPT}}$, that repeatedly invokes the (α, β)-approximation oracle over the available paths, using the true averages of the path parameters to select path subsets. As shown in [27], this algorithm achieves an asymptotic approximation ratio of $\frac{\alpha\beta}{1+\alpha\beta}$ with respect to the optimal dynamic policy Π^*, i.e., the expected cumulative reward of $\widetilde{\mathrm{OPT}}$ is $\frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T)$.

In the bandit setting, we adopt this ratio as a benchmark discount factor and define regret accordingly:

$$\mathrm{Reg}(T) = \left[\frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T) - \mathbb{E}\Big[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t} w_i r_i(t)\Big]\right]^+, \tag{8}$$

where $[\cdot]^+ = \max(\cdot, 0)$ is the positive-part operator, ensuring that only the shortfall relative to the discounted benchmark is counted; any episode in which the algorithm outperforms the benchmark contributes zero rather than negative regret. The first term is the discounted optimal expected reward, which serves as the performance benchmark; the second term is the expected cumulative reward actually obtained by the online algorithm.

To quantify how well the algorithm respects both constraints, we define the total long-term violation as the sum of the cost violation and the fidelity violation.

$$\begin{aligned} V(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T}\Big(\sum_{i\in\mathcal{P}_t}\mu_i(t) - B\Big)\right)^+\right] \\ + \mathbb{E}\left[\left(\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}[F_{\min} - F_i(t)]\right)^+\right], \end{aligned} \tag{9}$$

where the cost and fidelity terms both apply the positive-part operator to the horizon-wide cumulative sum, ensuring that the violation is never negative. We adopt this long-term definition rather than a per-time-slot guarantee because, in the unknown-distribution setting, the learner cannot identify a feasible action

at every time slot before gathering sufficient samples; per-time-slot feasibility would require prior knowledge of the true distributions that we do not assume.

The objective is to design an online algorithm that selects paths $\mathcal{P}_t$ for $t \in [T]$ and achieves sublinear regret, i.e., $\mathrm{Reg}(T) = o(T)$, while ensuring that the total violation is also sublinear, i.e., $V(T) = o(T)$. This implies that the average performance of the algorithm converges to that of $\widetilde{\mathrm{OPT}}$, and that both the average cost overrun and the average fidelity deficit vanish as T grows.

IV. Algorithm and Analysis

Algorithm 1 describes OQMPS in detail. Specifically, the algorithm begins with an initialization phase in which each path i is selected once to obtain initial empirical averages of its rate, cost, and fidelity. Its selection counter $n_i(t)$, which records the number of times path i has been selected up to time slot t, is initialized to $n_i(0) = 1$ (Lines 1–4). At time slot t, after observing the set of currently available (unblocked) paths $\mathcal{A}_t$ (Line 6), the algorithm computes a confidence radius $\mathrm{rad}_i(t)$ for each available path i, which depends on its selection count up to the previous slot (Line 8). Using this radius, it constructs optimistic upper-confidence-bound (UCB) estimates $\hat{r}_i(t)$ and $\hat{F}_i(t)$ for the rate and fidelity, respectively, together with a pessimistic lower-confidence-bound (LCB) estimate $\check{\mu}_i(t)$ for the cost (Lines 9–11). This encourages exploration of paths with potentially high rate, high fidelity, and low cost. The algorithm then invokes the (α, β)-approximation oracle over the available set $\mathcal{A}_t$ using these estimates. The oracle returns a subset $\mathcal{P}_t$ of exactly L paths that satisfies the estimated cost and fidelity constraints while approximately maximizing the estimated weighted rate (Line 13). Finally, after the paths are selected, the actual values of rate, cost, and fidelity are observed for each chosen path; the algorithm updates the corresponding empirical averages and increments the selection counters (Lines 14–20).

We now establish theoretical guarantees for the regret and constraint-satisfaction performance of OQMPS.

We first state a regret bound that characterizes the learning performance of the algorithm relative to an optimal dynamic policy. We then provide a constraint-violation bound that quantifies its ability to satisfy the per-slot cost budget and fidelity threshold in the long run.

Theorem 1. For OQMPS, the regret defined in (8) satisfies

$$\mathrm{Reg}(T) \leq \frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2+\pi^2 L) + \mathcal{O}(D_{\max}\cdot K). \tag{10}$$

Proof Sketch. The proof proceeds in three conceptual steps. First, the regret is reduced to the cumulative instantaneous regret—the per-slot loss incurred by selecting a suboptimal set due to estimation uncertainty—plus a blocking overhead that accounts for the temporary unavailability of selected paths. The blocking overhead, bounded by $\mathcal{O}(D_{\max}\cdot K)$, arises from the scheduling suboptimality under blocking constraints. Specifically, once a path is selected, it is forced into a recovery phase, forcing the algorithm to select alternative paths. While the dynamic optimal policy Π^* perfectly coordinates the expected selection rates to minimize this loss, our online algorithm—lacking perfect foresight of the recovery durations—may encounter unexpected blocking conflicts or deplete the high-rate paths within the recovery window. The cumulative performance gap caused by these suboptimal scheduling decisions scales linearly with the maximum delay $D_{\max}$ across the K paths, yielding the $\mathcal{O}(D_{\max}\cdot K)$ term.

Second, we decompose the instantaneous regret to handle the imperfect oracle and estimation uncertainty. When the oracle fails—with probability $1-\beta$—to return an α-approximate solution (i.e., a feasible subset whose weighted rate is at least an α fraction of the optimal weighted rate), the algorithm incurs an expected instantaneous loss. However, our benchmark is already discounted by $\frac{\alpha\beta}{1+\alpha\beta}$. Mathematically, the instantaneous regret can be decomposed into the α-scaled true maximum reward over the available paths, minus the expected reward obtained by the algorithm, plus a negative offset term proportional to $\alpha(\beta-1)$ times the true maximum reward. As proven in the Appendix, this specific discount factor acts as a negative offset that exactly cancels the expected loss caused

Algorithm 1 Online Quantum Multi-Path Selection (OQMPS)

Input: K, L, T, $f^{\alpha,\beta}$
Output: Set of selected paths $\mathcal{P}_t$ for each time slot t

1: **for** $i \in \{1,\dots,K\}$ **do**
2: Select path i and observe initial information at $t=0$
3: $n_i(0) \leftarrow 1$, $\bar{r}_i(0) \leftarrow r_i(0)$, $\bar{\mu}_i(0) \leftarrow \mu_i(0)$, $\bar{F}_i(0) \leftarrow F_i(0)$
4: **end for**
5: **for** $t = 1$ to T **do**
6: Obtain the set of available (unblocked) paths $\mathcal{A}_t$
7: **for** $i \in \mathcal{A}_t$ **do**
8: $\mathrm{rad}_i(t) \leftarrow \sqrt{\frac{3\ln t}{2n_i(t-1)}}$ ▷ Confidence radius
9: $\hat{r}_i(t) \leftarrow \bar{r}_i(t-1) + \mathrm{rad}_i(t)$ ▷ Rate estimate
10: $\check{\mu}_i(t) \leftarrow \bar{\mu}_i(t-1) - \mathrm{rad}_i(t)$ ▷ Cost estimate
11: $\hat{F}_i(t) \leftarrow \bar{F}_i(t-1) + \mathrm{rad}_i(t)$ ▷ Fidelity estimate
12: **end for**
▷ Call the (α,β)-approximation oracle in Definition 1
13: $\mathcal{P}_t \leftarrow f^{\alpha,\beta}(\mathcal{A}_t, \hat{\boldsymbol{r}}(t), \check{\boldsymbol{\mu}}(t), \hat{\boldsymbol{F}}(t))$
14: **for** $i \in \mathcal{P}_t$ **do**
15: Select path i for a connection request and observe $r_i(t)$, $\mu_i(t)$, $F_i(t)$
16: $n_i(t) \leftarrow n_i(t-1) + 1$
17: $\bar{r}_i(t) \leftarrow \frac{\bar{r}_i(t-1)n_i(t-1) + r_i(t)}{n_i(t)}$
18: $\bar{\mu}_i(t) \leftarrow \frac{\bar{\mu}_i(t-1)n_i(t-1) + \mu_i(t)}{n_i(t)}$
19: $\bar{F}_i(t) \leftarrow \frac{\bar{F}_i(t-1)n_i(t-1) + F_i(t)}{n_i(t)}$
20: **end for**
21: **end for**

by the oracle failure. Thus, the regret analysis reduces to the error caused by estimation uncertainty. By the Hoeffding inequality [30], the probability that the confidence bounds fail is at most $6K/t^3$. Summing these failure probabilities over T slots yields a cumulative loss of $\pi^2 KL$ for these tail events.

Third, on the good event where both the oracle succeeds and all confidence bounds hold, any selected suboptimal set has its suboptimality gap bounded by the sum of the confidence radii of its constituent paths. We split this gap using a sampling threshold $M = \sqrt{(48KL\ln T)/T}$. For gaps smaller than M, the total contribution is bounded by $T \cdot M = \mathcal{O}(\sqrt{KLT\ln T})$. For gaps larger than M, as a path is sampled repeatedly, its confidence radius shrinks at a rate of $\mathcal{O}(1/\sqrt{n_i(t)})$. Once the number of pulls exceeds a finite threshold, the radius shrinks below the gap, which prevents the path from incurring further regret; a standard counting argument then bounds the cumulative contribution by $\mathcal{O}(\sqrt{KLT\ln T})$. Combining these three parts yields the stated distribution-independent bound. The complete proof is provided in the Appendix. □

Theorem 2. For OQMPS, the total long-term violation $V(T)$ defined in (9) satisfies

$$V(T) \le 2\sqrt{24KLT\ln T} + 2\pi^2 KL + 2L\sqrt{T}. \tag{11}$$

Proof Sketch. The total long-term violation $V(T)$ is decomposed into the cost violation and the fidelity violation. At each time slot, the oracle guarantees that the estimated cost of the selected set does not exceed the budget, and that the estimated fidelity of every selected path is above the threshold. Therefore, the actual violations are bounded by the deviations between the true values and their estimates. Notably, unlike the regret bound, the violation bound does not depend on the approximation ratios α and β, because the oracle is invoked as a hard constraint solver: it ensures that the estimated parameters satisfy the constraints, regardless of how suboptimal the chosen set is in terms of reward maximization. Furthermore, the violation bound is independent of the maximum delay $D_{\max}$. This asymmetry arises because violation only measures the parameter estimation error of the paths at the moment they are selected, whereas blocking delays merely dictate when a path becomes available; they do not corrupt or shift the underlying statistical distribution of the path's rate, cost, or fidelity.

We decompose these deviations into three components. The first is the stochastic fluctuation of the realized values around their means. This forms a martingale difference sequence with bounded increments. By the Azuma-Hoeffding inequality, the expected positive part of these fluctuations accumulates at a rate of $\mathcal{O}(L\sqrt{T})$. The second component is the error between the empirical means and the true means. On the event where the confidence bounds hold, this estimation error is bounded by the sum of the confidence radii, which accumulates to $\mathcal{O}(\sqrt{KLT\ln T})$. Conversely, when the confidence bounds fail, the worst-case deviation per slot is bounded by L. Combined with the concentration failure probability of $6K/t^3$, the total additional violation accumulates to $\mathcal{O}(\pi^2 KL)$. The third component is the exploration bonus, which is exactly the sum of the confidence radii over all selected paths; this also accumulates to $\mathcal{O}(\sqrt{KLT\ln T})$ via a standard counting argument. By summing these three bounds for both the cost and fidelity violations and applying the positive-part operator, we obtain the stated sublinear total violation bound. The complete proof is provided in the Appendix. □

V. Experiments

A. Experiment Setup

1) Parameter Setting: We simulate a quantum network with $K = 50$ precomputed end-to-end paths over a horizon of $T = 5000$ time slots. At each time slot, the network selects $L = 5$ paths from the available set $\mathcal{A}_t$ ($|\mathcal{A}_t| > L$) to serve concurrent connection requests. Results are averaged over 40 independent runs.

For each path i, the priority weight w_i is independently drawn from a uniform distribution over $(0, 1]$. Instantaneous rate, cost, and fidelity are sampled from truncated normal distributions with standard deviation $\sigma = 0.08$ and support $(0, 1)$. The fidelity threshold is set to $F_{\min}$, and the cost budget is $B = B_{\text{scale}} \cdot \frac{L}{K} \sum_{i=1}^{K} \mu_i$, where B_{scale} is a budget-scaling factor that controls the tightness of the cost constraint. Blocking delays follow a uniform distribution whose mean d_i is drawn from $\text{Unif}(0, 5)$ for each path, with $D_{\max} = \max_i 2d_i$; geometric and exponential delay distributions are also examined for robustness.

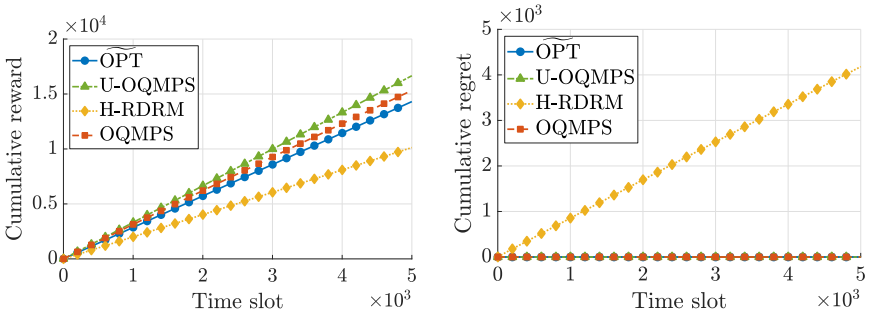
2) Baseline Algorithms: We implement the (α, β)-approximation oracle in OQMPS using an FPTAS with approximation parameter $\epsilon = 0.01$, i.e., $\alpha = 0.99$ and $\beta = 1$. We compare OQMPS against the following baselines:

- **$\widetilde{\text{OPT}}$**: An ideal full-information benchmark that uses the same approximation oracle with the same parameters as OQMPS. It has full knowledge of all path parameters and selects the feasible L paths that approximately maximize total weighted rate.
- **U-OQMPS** (Unconstrained OQMPS): A variant of OQMPS that optimizes the rate alone while ignoring the cost and fidelity constraints.
- **H-RDRM** (Heuristic Rate Delay Ratio Maximization): A heuristic that selects paths maximizing $\sum w_i \bar{r}_i / \bar{d}_i$ based on empirical averages. We consider this heuristic because of the blocking effect: a high-rate path with an excessively long blocking delay may contribute less to the long-term average reward than a lower-rate path that is available more frequently. H-RDRM therefore balances immediate reward against frequent availability.

Performance is evaluated using three metrics: cumulative reward, regret against $\widetilde{\text{OPT}}$, and total long-term violation.

B. Cumulative Reward and Regret

We first evaluate the four algorithms under the default configuration, where blocking delays follow a uniform distribution, the cost budget scaling factor is set to $B_{\text{scale}} = 1.0$, and the fidelity threshold is $F_{\min} = 0.7$.



(a) Cumulative Reward (b) Cumulative Regret

Fig. 2. Comparison of cumulative reward and cumulative regret.

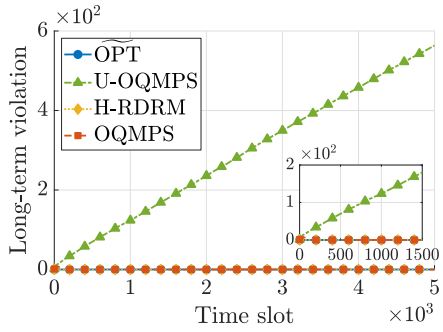


Fig. 3. Comparison of long-term violation.

Fig. 2(a) presents cumulative reward over time. U-OQMPS achieves the highest cumulative reward because it greedily maximizes the rate using UCB estimates while disregarding all constraints. Our proposed OQMPS attains the second-highest reward and clearly outperforms H-RDRM. Notably, OQMPS surpasses the ideal benchmark $\widetilde{\text{OPT}}$. This is because $\widetilde{\text{OPT}}$ satisfies constraints more strictly in expectation, whereas OQMPS, as an online learning algorithm, explores slightly more aggressively and may incur marginal short-term violations. This trade-off enables OQMPS to achieve a higher realized reward than the ideal but conservative $\widetilde{\text{OPT}}$ benchmark without compromising long-term constraint safety.

Fig. 2(b) shows the cumulative regret relative to $\widetilde{\text{OPT}}$. OQMPS achieves near-zero regret, indicating effective learning and convergence to near-optimal decisions under the given constraints. U-OQMPS also exhibits a near-zero regret curve, but this does not imply convergence to a truly optimal feasible solution; rather, it reflects convergence to the unconstrained optimum, which violates both cost and fidelity constraints, as evidenced by its severe violation in Fig. 3. In contrast, H-RDRM exhibits approximately linear regret growth. The inferior performance of H-RDRM stems from its objective of maximizing the weighted rate-to-delay ratio $w_i r_i / d_i$ rather than the rate itself.

1) Constraint Violation: Fig. 3 reports total long-term violation, which accumulates the positive parts of cost overruns and fidelity shortfalls. U-OQMPS shows a continuous, nearly linear increase in violation over the entire horizon. This indicates that unconstrained rate maximization comes at the expense of resource-budget exhaustion and frequent selection of paths below the fidelity threshold.

In contrast, the violation of OQMPS remains near zero throughout the horizon, exhibiting only a negligible, short-lived fluctuation during the initial exploration phase. This fluctuation results from the limited observations available initially. As the confidence bounds tighten with more observations, the algorithm successfully respects both constraints in the long run. H-RDRM also exhibits near-zero violations in both metrics, despite its poor reward performance. This is because its empirical-mean-based decisions, while suboptimal for reward maximization, tend to be inherently conservative, thereby avoiding high-cost or low-fidelity paths.

These results collectively show that OQMPS achieves the highest reward among all constraint-aware algorithms while maintaining near-zero long-term violation, consistent with the theoretical analysis in Sec. IV. The UCB-based optimistic estimation for rates enables effective exploration, while the LCB-based pessimistic estimation for costs provides reliable budget protection, and the UCB-based filtering for fidelity ensures that only high-fidelity paths enter the candidate set.

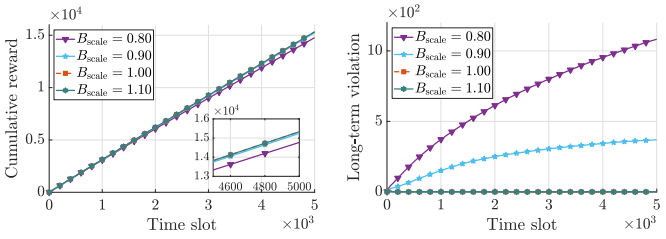
2) Impact of Parameters: We next evaluate the robustness of OQMPS under varying configurations, focusing on three parameters: the cost budget scaling factor B_{scale}, the fidelity threshold $F_{\min}$, and the delay distribution.

Impact of cost budget: We first vary the cost budget scaling factor $B_{\text{scale}} \in \{0.8, 0.9, 1.0, 1.1\}$. Fig. 4(a) and Fig. 4(b) report the cumulative reward and the total long-term violation, respectively.

As shown in Fig. 4(a), the cumulative reward increases monotonically with B_{scale}. Moreover, the performance gap across configurations remains moderate: even under the tightest budget, OQMPS attains substantial reward, indicating effective resource allocation. The inset figure confirms that the relative ordering stabilizes over time.

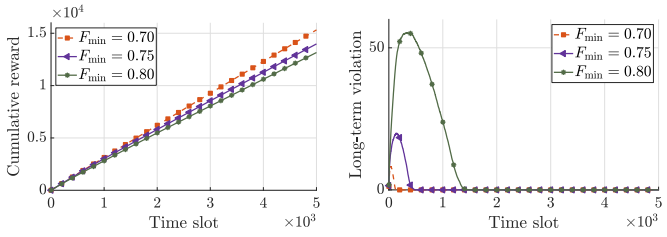
Fig. 4(b) shows that total long-term violation is highly sensitive to B_{scale}. For the tightest budget $B_{\text{scale}} = 0.8$, the violation rises persistently over the entire horizon, accumulating to the largest magnitude among all settings. When the budget is relaxed to $B_{\text{scale}} = 0.9$, the violation drops substantially compared with the 0.8 case, yet a non-negligible amount of cumulative violation still remains. In contrast, for $B_{\text{scale}} \geq 1.0$, the violation is effectively suppressed to near zero, with virtually no long-term accumulation. Importantly, even under the tightest budget, the violation growth remains sublinear, consistent with our theoretical guarantee in Sec. IV.

The total violation observed under tight budgets is primarily driven by cost overruns because the fidelity constraint is largely decoupled from the cost budget. The UCB-based high-fidelity filtering mechanism operates independently, ensuring path quality even when the algorithm struggles to meet the cost limit. Accordingly, total violation for $B_{\text{scale}} = 0.8$ and 0.9 accumulates persistently because cost violations dominate,



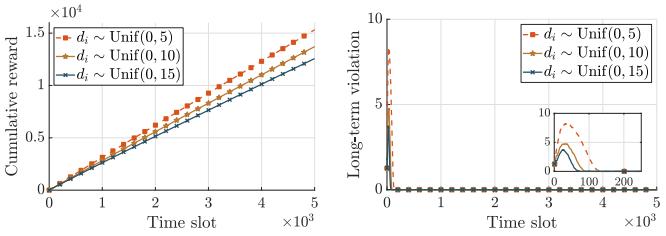
(a) Cumulative reward (b) Long-term violation

Fig. 4. Impact of cost budget.



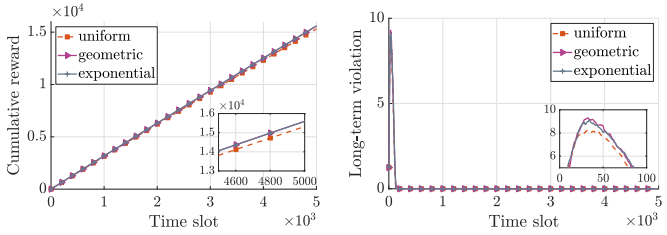
(a) Cumulative reward (b) Long-term violation

Fig. 5. Impact of fidelity threshold.



(a) Cumulative reward (b) Long-term violation

Fig. 6. Impact of mean blocking delays.



(a) Cumulative reward (b) Long-term violation

Fig. 7. Impact of delay distributions.

whereas for $B_{\text{scale}} \geq 1.0$, both cost and fidelity violations are controlled.

In summary, OQMPS maintains stable performance across budget settings: reward increases predictably as the budget is relaxed, total constraint violation remains sublinear and vanishes for $B_{\text{scale}} \geq 1.0$, and fidelity guarantees are preserved regardless of budget scaling.

Impact of fidelity threshold: We then vary the fidelity threshold $F_{\min} \in \{0.70, 0.75, 0.80\}$. Fig. 5(a) and Fig. 5(b) report the cumulative reward and the total long-term violation.

As shown in Fig. 5(a), the cumulative reward decreases monotonically as $F_{\min}$ increases. This occurs because a stricter fidelity threshold shrinks the set of feasible paths. Consequently, the algorithm has fewer options for selecting high-rate paths, which leads to a lower overall reward. The strictest threshold yields the lowest reward, while the most lenient achieves the highest.

Fig. 5(b) presents total long-term violation under different fidelity thresholds. The algorithm's behavior varies with the strictness of the threshold. For the most lenient threshold $F_{\min} = 0.70$, the violation remains virtually zero throughout the horizon, with no noticeable spike. As the threshold increases to $F_{\min} = 0.75$, a slight transient fluctuation appears during the early exploration phase but decays rapidly to zero. The strictest threshold $F_{\min} = 0.80$ exhibits a more visible initial spike and the slowest convergence. This inverse relationship arises because a higher threshold reduces the number of paths that naturally satisfy the fidelity requirement. Consequently, regardless of the selection policy, a chosen set is more likely to include paths whose true fidelity falls below the threshold during the initial learning phase, leading to larger and more persistent spikes. Importantly, for all three settings, total long-term violation converges to zero, demonstrating the robustness of the algorithm across fidelity requirements.

Impact of mean blocking delay: We investigate how the mean blocking delay d_i affects the performance of OQMPS by varying the upper bound of the distribution from which d_i is drawn, i.e., $d_i \sim \text{Unif}(0, 5)$, $\text{Unif}(0, 10)$, and $\text{Unif}(0, 15)$.

As shown in Fig. 6(a), cumulative reward increases approximately linearly over time for all three settings, with a clear ordering: $d_i \sim \text{Unif}(0, 5)$ achieves the highest reward, followed by $d_i \sim \text{Unif}(0, 10)$, whereas $d_i \sim \text{Unif}(0, 15)$ yields the lowest. This is because shorter blocking delays allow paths with high optimistic rate estimates to be selected more frequently, thereby contributing more to the long-term average rate. The gap between the three curves widens gradually over time, and no convergence or crossover is observed.

Fig. 6(b) presents total long-term violation under the same three delay settings. All violation curves remain extremely close to zero throughout the horizon, with no pronounced spikes or persistent accumulation. Moreover, the differences in violation across different d_i values are almost negligible. This indicates that the delay magnitude does not substantially affect the ability of the algorithm to respect the constraints. The constraint-enforcement mechanism of OQMPS is largely insensitive to the delay statistics and successfully maintains near-zero long-term violation regardless of the mean blocking delay.

Impact of blocking delay distributions: We further test the robustness of OQMPS under different delay distributions, including uniform, geometric, and exponential distributions.

As shown in Fig. 7(a), cumulative reward is slightly higher under geometric and exponential delay distributions than under the uniform distribution.

Fig. 7(b) presents total long-term violation under the three delay distributions. Consistent with the results in Fig. 6(b),

the violation curves for all three distributions remain near zero throughout the horizon. No pronounced spikes or persistent accumulation are observed, and the differences among the three curves are negligible. This result shows that the constraint-satisfaction performance of the algorithm is largely insensitive to the shape of the delay distribution. Regardless of whether the delay is light-tailed or heavy-tailed, OQMPS learns to satisfy both the cost and fidelity constraints in the long run.

VI. Conclusion & Future Work

This paper investigates online multi-path selection in quantum networks, in which an algorithm must learn the unknown distributions of rate, cost, and fidelity while satisfying a per-slot cost budget and a fidelity threshold. We propose an online learning algorithm that maintains confidence bounds for all path statistics, constructs optimistic estimates for rate and fidelity and a pessimistic estimate for cost, and invokes an (α, β)-approximation oracle to select a feasible subset of paths. The theoretical analysis establishes sublinear regret and sublinear total long-term violation. Simulations on a 50-path quantum network show that OQMPS achieves reward performance competitive with the ideal $\widetilde{\text{OPT}}$ benchmark while maintaining near-zero total long-term violation.

Future work includes eliminating the reliance on precomputed end-to-end paths, extending the framework to accommodate multiple resource types, explicitly incorporating quantum-memory coherence times into the blocking model, and exploring adaptive oracle-precision tuning to further improve regret bounds in practical deployments.

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Appendix

A. Preliminaries for Regret Analysis

We first establish structural properties of the optimal dynamic policy under stochastic blocking and then introduce the decomposition framework that connects the regret to the per-slot instantaneous loss.

Lemma 1. For any path i with expected blocking delay $d_i = \mathbb{E}[D_i(t)]$ and maximum delay $D_{\max}$, we have:

$$z_i \leq \frac{1}{d_i + 1} + \mathcal{O}\left(\frac{D_{\max}}{T}\right).$$

Proof. We denote by $\mathcal{P}_t^*$ and $\mathcal{B}_t^*$ the sets of selected and blocked paths of the optimal online policy π^*, respectively, at any time t. The event that path i is selected at time slot t and the event that i is blocked at t are mutually exclusive, namely, $\mathbb{I}(i \in \mathcal{P}_t^*) + \mathbb{I}(i \in \mathcal{B}_t^*) \leq 1$. Further, a path i is blocked at t if and only if it has been played in some previous time slot and is still unavailable at t. Therefore, at any time slot t, we get:

$$\mathbb{I}(i \in \mathcal{P}_t^*) + \sum_{t'<t} \mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t') \leq 1.$$

Taking expectation in the above expression, we obtain:

$$\mathbb{P}(i \in \mathcal{P}_t^*) + \sum_{t'<t} \mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t')] \leq 1, \ \forall t \tag{12}$$

The events $\{i \in \mathcal{P}_{t'}^*\}$ and $\{D_i(t') \geq t - t'\}$ are conditionally independent given the history of paths selected up to and including time $t' - 1$, which we denote by $H_{t'-1}^*$; hence:

$$\begin{aligned}
&\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t')] \\
&= \mathbb{E}\big[\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*)\mathbb{I}(D_i(t') \geq t - t') \mid H_{t'-1}^*]\big] \\
&= \mathbb{E}\big[\mathbb{E}[\mathbb{I}(i \in \mathcal{P}_{t'}^*) \mid H_{t'-1}^*]\mathbb{E}[\mathbb{I}(D_i(t') \geq t - t') \mid H_{t'-1}^*]\big] \\
&= \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t')
\end{aligned}$$

Summing (12) over all time slots t and using $\mathbb{P}(D_i(t) \geq 0) = 1$, we obtain:

$$\sum_{t=1}^{T} \sum_{t' \leq t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t') \leq T. \tag{13}$$

Since the variables $D_i(t)$ are i.i.d., for any sample D_i from the delay distribution of path i we have $\mathbb{P}(D_i \geq t - t') = \mathbb{P}(D_i(t) \geq t - t')$. Moreover, since D_i is supported on $\{0, 1, ..., D_{\max}\}$, we can write:

$$\begin{aligned}
&\sum_{t=1}^{T} \sum_{t' \leq t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i(t') \geq t - t') \\
&= \sum_{t=1}^{T} \sum_{t'=\max\{1, t-D_{max}\}}^{t} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \sum_{t=t'}^{\min\{T, t'+D_{\max}\}} \mathbb{P}(i \in \mathcal{P}_{t'}^*)\mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) \sum_{t=t'}^{\min\{T, t'+D_{\max}\}} \mathbb{P}(D_i \geq t - t') \\
&\geq \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) \sum_{t=t'}^{t'+D_{\max}} \mathbb{P}(D_i \geq t - t') \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \\
&= \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*)(1 + \mathbb{E}[D_i]) \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t'),
\end{aligned}$$

Therefore, inequality (13) becomes:

$$\begin{aligned}
&\sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*)(1 + \mathbb{E}[D_i]) \\
&\quad - \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \leq T.
\end{aligned}$$

Finally, by rearranging terms in the above expression and dividing by $(\mathbb{E}[D_i] + 1) \cdot T$, we conclude that:

$$\begin{aligned}
\frac{1}{T} \sum_{t'=1}^{T} \mathbb{P}(i \in \mathcal{P}_{t'}^*) &\leq \frac{1}{(\mathbb{E}[D_i] + 1)} \\
&\quad + \frac{1}{T(\mathbb{E}[D_i] + 1)} \sum_{t'=T-D_{\max}+1}^{T} \sum_{t=t'}^{T} \mathbb{P}(D_i \geq t - t') \\
&\leq \frac{1}{(\mathbb{E}[D_i] + 1)} + \frac{D_{\max}(\mathbb{E}[D_i] + 1)}{T(\mathbb{E}[D_i] + 1)} \\
&= \frac{1}{(\mathbb{E}[D_i] + 1)} + \frac{D_{\max}}{T}.
\end{aligned}$$

□

Lemma 2. For any subset of paths $\mathcal{S} \subseteq [K]$, with $\mathfrak{P}(\mathcal{S})$ the feasible family defined in Definition 1, we have:

$$\max_{\mathcal{Q} \in \mathfrak{P}(\mathcal{S})} \sum_{i \in \mathcal{Q}} w_i r_i \geq \sum_{i \in \mathcal{S}} w_i r_i z_i$$

Proof. At any time t, we have:

$$\max_{\mathcal{Q} \subseteq \mathcal{S}} \sum_{i \in \mathcal{Q}} w_i r_i \geq \sum_{i \in \mathcal{S} \cap \mathcal{P}_{t'}^*} w_i r_i = \sum_{i \in \mathcal{S}} w_i r_i \mathbb{I}(i \in \mathcal{P}_{t'}^*).$$

By averaging over time and taking expectations, we conclude that:

$$\max_{\mathcal{Q}\subseteq\mathcal{S}} \sum_{i\in\mathcal{Q}} w_i r_i \geq \frac{1}{T}\sum_{t=1}^{T}\sum_{i\in\mathcal{S}} w_i r_i \mathbb{P}(i\in\mathcal{P}_{t'}^*) = \sum_{i\in\mathcal{S}} w_i r_i z_i.$$

□

Definition 2 (Instantaneous regret)**.** At time t, the instantaneous regret of the algorithm is defined as:

$$\Gamma_t = \alpha\beta \cdot \max_{\mathcal{Q}\in\mathfrak{P}(\mathcal{A}_t)} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i,$$

where $\mathcal{A}_t$ is the set of available paths at the beginning of time slot t, $\mathfrak{P}(\mathcal{A}_t)$ is the family of feasible subsets of $\mathcal{A}_t$, and $\mathcal{P}_t$ is the set selected by the algorithm. Intuitively, Γ_t measures the gap between an $\alpha\beta$-fraction of the optimal value over the available paths and the actual reward obtained.

Lemma 3. The regret of the algorithm satisfies:

$$\mathrm{Reg}(T) \leq \frac{1}{1+\alpha\beta}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right] + \mathcal{O}(D_{\max}\cdot K).$$

Proof. Let H_t be the history of path selections made by our bandit algorithm up to and including time t. We define the total expected reward collected by our algorithm over T time slots as:

$$\begin{aligned}
\left[\sum_{t=1}^{T} r(t)\right] &= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)\right] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}[w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}\big[\mathbb{E}[w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)|H_{t-1}]\big] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}\big[\mathbb{E}[w_i r_i(t)|H_{t-1}]\mathbb{E}[\mathbb{I}(i\in\mathcal{P}_t)|H_{t-1}]\big] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K}\mathbb{E}[w_i r_i\mathbb{I}(i\in\mathcal{P}_t)] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right],
\end{aligned}$$

where we use the fact that, at each time slot t, the reward $w_i r_i$ and the choice of the set $\mathcal{P}_t$ are independent conditioned on the history H_{t-1}.

We recall the expected pulling rate of each path:

$$z_i = \mathbb{E}\left[\frac{1}{T}\sum_{t=1}^{T}\mathbb{I}(i\in\mathcal{P}_t^*)\right]$$

where $\mathcal{P}_t^*$ is the set of paths selected by the optimal online policy π^* at t.

Applying the definition of Γ_t at every time slot t gives:

$$\sum_{i\in\mathcal{P}_t} w_i r_i = \alpha\beta\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i - \Gamma_t.$$

where $\mathcal{A}_t$ is the set of available paths for our algorithm at the beginning of time slot t, and $\mathcal{Q}$ is a feasible subset satisfying the cardinality, cost, and fidelity constraints.

For any t, we have:

$$\begin{aligned}
\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right] &= \alpha\beta\cdot\mathbb{E}\left[\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i\right] - \mathbb{E}[\Gamma_t] \\
&\geq \alpha\beta\cdot\mathbb{E}\left[\sum_{i\in\mathcal{A}_t} w_i r_i z_i\right] - \mathbb{E}[\Gamma_t] \\
&= \alpha\beta\cdot\sum_{i=1}^{K} w_i r_i z_i - \alpha\beta\cdot\sum_{i\in\mathcal{B}_t} w_i r_i z_i - \mathbb{E}[\Gamma_t]
\end{aligned}$$

where $\mathcal{B}_t$ is the set of blocked (unavailable) paths of our algorithm. The inequality follows from Lemma 2.

Note that for any t and any path $i\in\mathcal{B}_t$, we have:

$$\mathbb{I}(i\in\mathcal{B}_t) = \sum_{t'<t}\mathbb{I}(i\in\mathcal{P}_{t'})\mathbb{I}(D_i(t')\geq t-t').$$

Using this fact, we have:

$$\begin{aligned}
&\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{B}_t} w_i r_i z_i\right] \\
&= \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i z_i \sum_{t'<t}\mathbb{I}(i\in\mathcal{P}_{t'})\mathbb{I}(D_i(t')\geq t-t')\right] \\
&= \sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i \sum_{t'<t}\mathbb{P}(i\in\mathcal{P}_{t'})\mathbb{P}(D_i(t')\geq t-t') \\
&= \sum_{t'=1}^{T}\sum_{i=1}^{K}\mathbb{P}(i\in\mathcal{P}_{t'}) w_i r_i z_i \sum_{t>t'}\mathbb{P}(D_i(t')\geq t-t') \\
&\leq \sum_{t'=1}^{T}\sum_{i=1}^{K}\mathbb{P}(i\in\mathcal{P}_{t'}) w_i r_i z_i d_i \\
&\leq \left[\sum_{t=1}^{T} r(t)\right] + \mathcal{O}(D_{\max}\cdot K)
\end{aligned} \tag{14}$$

By summing over t, we have:

$$\begin{aligned}
&\left[\sum_{t=1}^{T} r(t)\right] = \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i=1}^{K} w_i r_i(t)\mathbb{I}(i\in\mathcal{P}_t)\right] \\
&= \sum_{t=1}^{T}\mathbb{E}\left[\sum_{i\in\mathcal{P}_t} w_i r_i\right] \\
&\geq \alpha\beta\mathrm{Rew}^*(T) - \alpha\beta\left[\sum_{t=1}^{T} r(t)\right] - \mathcal{O}(D_{\max}\cdot K) - \mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right].
\end{aligned}$$

Rearranging the expression above, we obtain:

$$\begin{aligned}
\left[\sum_{t=1}^{T} r(t)\right] \geq \frac{\alpha\beta}{1+\alpha\beta}\mathrm{Rew}^*(T) - \mathcal{O}(D_{\max}\cdot K) \\
- \frac{1}{1+\alpha\beta}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right]
\end{aligned}$$

□

Lemma 4. Let $\mathcal{N}_t$ denote the event that for all paths $i \in [K]$, the following inequalities hold simultaneously: $|\bar{r}_i(t-1) - r_i| \le \mathrm{rad}_i(t)$, $|\bar{\mu}_i(t-1) - \mu_i| \le \mathrm{rad}_i(t)$, $|\bar{F}_i(t-1) - F_i| \le \mathrm{rad}_i(t)$, where $\mathrm{rad}_i(t) = \sqrt{\frac{3\ln t}{2n_i(t-1)}}$. Then we have $\mathbb{P}(\mathcal{N}_t) \ge 1 - \frac{6K}{t^3}$.

Proof. For any $n_i(t-1) > 0$ and $i \in [K]$, using Hoeffding's inequality and the fact that $n_i(t-1) \le t-1$, we have that:

$$\begin{aligned}
&\mathbb{P}\left(|\bar{r}_i(t-1) - r_i| \ge \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \\
&= \sum_{s=1}^{t-1} \mathbb{P}(n_i(t-1) = s) \cdot \mathbb{P}\left(|\bar{r}_i(t-1) - r_i| \ge \sqrt{\frac{3\ln t}{2s}} \,\middle|\, n_i(t-1) = s\right) \\
&\le \sum_{s=1}^{t-1} \mathbb{P}(n_i(t-1) = s) \cdot 2e^{-3\ln t} \\
&= 2t^{-3} \sum_{s=1}^{t-1} \mathbb{P}(n_i(t-1) = s) \\
&\le \frac{2}{t^3}
\end{aligned}$$

We also have that:

$$\mathbb{P}\left(|\bar{\mu}_i(t-1) - \mu_i| \ge \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \le \frac{2}{t^3}$$
$$\mathbb{P}\left(|\bar{F}_i(t-1) - F_i| \ge \sqrt{\frac{3\ln t}{2n_i(t-1)}}\right) \le \frac{2}{t^3}$$

The lemma follows by taking the union bound over the three variables and over all K paths:

$$\begin{aligned}
&\mathbb{P}(\neg\mathcal{N}_t) = \mathbb{P}\Bigg(\exists i \in [K] : |\bar{r}_i(t-1) - r_i| > \mathrm{rad}_i(t) \text{ or } \\
&|\bar{\mu}_i(t-1) - \mu_i| > \mathrm{rad}_i(t) \text{ or } |\bar{F}_i(t-1) - F_i| > \mathrm{rad}_i(t)\Bigg) \\
&\le 3 \cdot \sum_{i=1}^{K} \frac{2}{t^3} \le \frac{6K}{t^3}.
\end{aligned}$$

□

B. Proof of Theorem 1

With the above definitions and lemmas in place, we now present the complete proof of the regret bound.

Theorem 1. For OQMPS, the regret defined in (8) satisfies

$$\begin{aligned}
\mathrm{Reg}(T) \le &\frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2+\pi^2 L) \\
&+ \mathcal{O}(D_{\max} \cdot K).
\end{aligned}$$

Proof. According to Lemma 3, to bound the regret of our bandit algorithm, it suffices to bound the cumulative instantaneous regret. This loss can be written as:

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T} \Gamma_t\right] &= \mathbb{E}\left[\sum_{t=1}^{T} \left(\alpha\beta \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i\right)\right] \\
&= \mathbb{E}\Bigg[\sum_{t=1}^{T} \Bigg(\alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i \\
&\qquad + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Bigg)\Bigg].
\end{aligned}$$

Focusing on the term $\alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i$ of the loss, we define the bad set and the suboptimality gap to characterize the decisions that contribute to this quantity:

Definition 3 (Bad set)**.** A subset $\mathcal{S} \subseteq \mathcal{A}$ is called a bad set with respect to the available set $\mathcal{A}$ if: $\sum_{i\in\mathcal{S}} w_i r_i \le \alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}} \sum_{i\in\mathcal{Q}} w_i r_i$. Moreover, we define the bad sets of the available set $\mathcal{A}$ as:

$$\mathcal{S}_B(\mathcal{A}) = \left\{\mathcal{S} \,\middle|\, \sum_{i\in\mathcal{S}} w_i r_i \le \alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}} \sum_{i\in\mathcal{Q}} w_i r_i\right\}$$

where $|\mathcal{S}| = L$, and $\mathcal{Q}$ is a feasible subset satisfying the cardinality, cost, and fidelity constraints.

Definition 4 (Suboptimality gap)**.** The suboptimality gap of a bad set $\mathcal{S}$ with respect to $\mathcal{A}$ is defined as:

$$\Delta(\mathcal{S}, \mathcal{A}) = \alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{S}} w_i r_i.$$

where $|\mathcal{S}| = L$, and $\mathcal{Q}$ is a feasible subset satisfying the cardinality, cost, and fidelity constraints.

Let $\Delta_{\max} = \max_{\mathcal{S}\in\mathcal{S}_B(\mathcal{A}), \mathcal{A}\subseteq\{1,\dots,K\}} \Delta(\mathcal{S}, \mathcal{A})$ denote the maximum possible gap.

Thus, we can write:

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T} \Gamma_t\right] &= \mathbb{E}\Bigg[\sum_{t=1}^{T} \Bigg(\alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i - \sum_{i\in\mathcal{P}_t} w_i r_i \\
&\qquad + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Bigg)\Bigg] \\
&\le \mathbb{E}\Bigg[\sum_{t=1}^{T} \Bigg(\mathbb{I}(\mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t))(\alpha \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i \\
&\quad - \sum_{i\in\mathcal{P}_t} w_i r_i) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Bigg)\Bigg] \\
&= \mathbb{E}\Bigg[\sum_{t=1}^{T} \Bigg(\mathbb{I}(\mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t))\Delta(\mathcal{P}_t, \mathcal{A}_t) \\
&\qquad + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Bigg)\Bigg]
\end{aligned} \tag{15}$$

We now treat the parts of the regret resulting from oracle failures and from bad sampling separately. Let $\mathcal{O}_t$ denote the

event that the oracle succeeds at time slot t. We decompose the indicator in (15) based on whether the events $\mathcal{O}_t$ and $\mathcal{N}_t$ hold, as follows:

$$(15) \leq \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{I}(\mathcal{O}_t, \mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t)) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\right] \quad (16)$$

$$+ \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\neg\mathcal{O}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t)) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Bigg] \quad (17)$$

$$+ \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{I}(\neg\mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t)) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\right] \quad (18)$$

We bound each of the above terms separately. For (17), because the available set changes at each slot, we consider the suboptimality gap of the worst solution in the available set $\mathcal{F}$, denoted $\Delta_{\max}(\mathcal{F})$:

$$\Delta_{\max}(\mathcal{F}) = \max_{\mathcal{S}\in\mathcal{S}_B} \Delta(\mathcal{S}, \mathcal{F}).$$

We use this definition to upper-bound the suboptimality gap appearing in (17):

$$\begin{aligned}
(17) &\leq \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}\Big((\neg\mathcal{O}_t) \cdot \Delta_{\max}(\mathcal{A}_t) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Big)\Bigg] \\
&= \mathbb{E}\Bigg[\sum_{t=1}^{T} \sum_{\mathcal{F}\subseteq[K]} \mathbb{E}\Big[\mathbb{I}(\neg\mathcal{O}_t) \cdot \Delta_{\max}(\mathcal{F}) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i \Big| \mathcal{A}_t = \mathcal{F}\Big]\mathbb{I}(\mathcal{A}_t = \mathcal{F})\Bigg] \\
&= \mathbb{E}\Bigg[\sum_{t=1}^{T} \sum_{\mathcal{F}\subseteq[K]} \Big(\mathbb{P}(\neg\mathcal{O}_t|\mathcal{A}_t = \mathcal{F}) \cdot \Delta_{\max}(\mathcal{F}) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Big)\mathbb{I}(\mathcal{A}_t = \mathcal{F})\Bigg] \\
&\leq \mathbb{E}\Bigg[\sum_{t=1}^{T} \sum_{\mathcal{F}\subseteq[K]} \Big((1-\beta) \cdot \Delta_{\max}(\mathcal{F}) + \alpha(\beta-1) \cdot \max_{\mathcal{Q}\subseteq\mathcal{A}_t} \sum_{i\in\mathcal{Q}} w_i r_i\Big)\mathbb{I}(\mathcal{A}_t = \mathcal{F})\Bigg] \leq 0
\end{aligned} \quad (19)$$

For the term (18), using the definition of $\Delta_{\max}$ and Lemma 4, we have:

$$\begin{aligned}
(18) &= \mathbb{E}\left[\sum_{t=1}^{T} \mathbb{I}(\neg\mathcal{N}_t) \cdot \Delta_{\max}\right] = \sum_{t=1}^{T} \mathbb{P}(\neg\mathcal{N}_t) \cdot \Delta_{\max} \\
&\leq \sum_{t=1}^{T} \frac{6K}{t^3} \Delta_{\max} \leq \sum_{t=1}^{T} \frac{6K}{t^2} \Delta_{\max} \leq \pi^2 K \Delta_{\max} \leq \pi^2 K L.
\end{aligned} \quad (20)$$

To bound (16), we first define the sampling threshold for a given gap level M:

Definition 5 (Sampling threshold)**.** We define the sampling threshold as:

$$l_T(M) = \frac{24L^2 \ln T}{M^2},$$

We also define the following function:

$$\kappa_T(M, s) = \begin{cases} 2, & s = 0, \\ \sqrt{\frac{24 \ln T}{s}}, & 1 \leq s \leq l_T(M), \\ 0, & s > l_T(M). \end{cases}$$

We define $M = \sqrt{(48KL \ln T)/T}$. We use this definition to rewrite the term (16) as:

$$(16) = \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\mathcal{O}_t, \mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t), \Delta(\mathcal{P}_t, \mathcal{A}_t) \geq M) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\Bigg] \quad (21)$$

$$+ \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\mathcal{O}_t, \mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t), \Delta(\mathcal{P}_t, \mathcal{A}_t) < M) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\Bigg]. \quad (22)$$

The second term of the above equation can be bounded using the fact that $\Delta(\mathcal{P}_t, \mathcal{A}_t) < M$, as follows:

$$\begin{aligned}
(22) &\leq \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t), \Delta(\mathcal{P}_t, \mathcal{A}_t) < M) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\Bigg] \leq T \cdot M = T\sqrt{(48KL \ln T)/T} \\
&= \sqrt{48LKT \ln T}.
\end{aligned} \quad (23)$$

The first term can be bounded using Lemma 5:

$$\begin{aligned}
(21) &= \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\mathcal{O}_t, \mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t), \Delta(\mathcal{P}_t, \mathcal{A}_t) \geq M) \cdot \Delta(\mathcal{P}_t, \mathcal{A}_t)\Bigg] \\
&\leq \mathbb{E}\Bigg[\sum_{t=1}^{T} \mathbb{I}(\mathcal{O}_t, \mathcal{N}_t, \mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t), \Delta(\mathcal{P}_t, \mathcal{A}_t) \geq M) \cdot \kappa_T(M, n_i(t-1))\Bigg] \\
&\leq \mathbb{E}\left[\sum_{i=1}^{K} \sum_{s=0}^{n_i(T)} \kappa_T(M, s)\right] \\
&\leq 2K + \sum_{i=1}^{K} 48 \frac{L}{M} \ln T \\
&= 2K + \sqrt{48KLT \ln T}
\end{aligned} \quad (24)$$

We conclude our proof by combining the above (19), (20), (23) and (24) with Lemma 3:

$$\begin{aligned}\mathrm{Reg}(T) &\le \frac{1}{1+\alpha\beta}\mathbb{E}\left[\sum_{t=1}^{T}\Gamma_t\right] + \mathcal{O}(D_{\max}\cdot K)\\ &\le \frac{1}{1+\alpha\beta}\left(\pi^2 KL + 2\sqrt{48KLT\ln T} + 2K\right)\\ &\quad + \mathcal{O}(D_{\max}\cdot K)\\ &\le \frac{14\sqrt{KLT\ln T}}{1+\alpha\beta} + \frac{K}{1+\alpha\beta}(2+\pi^2 L)\\ &\quad + \mathcal{O}(D_{\max}\cdot K).\end{aligned}$$

□

Lemma 5. For any time t, if $\mathcal{O}_t$ and $\mathcal{N}_t$ both hold, the algorithm selects a bad set $\mathcal{P}_t \in \mathcal{S}_B(\mathcal{A}_t)$, and $\Delta(\mathcal{P}_t, \mathcal{A}_t) \ge M$, then:

$$\Delta(\mathcal{P}_t, \mathcal{A}_t) \le \sum_{i\in\mathcal{A}_t} \kappa_T(M, n_i(t-1)).$$

Proof. By the definition of $\mathcal{N}_t$, if $\mathcal{N}_t$ holds at time slot t, then $\forall i$ the optimistic and pessimistic estimates are bounded as follows:

$$|\hat{r}_i(t) - r_i| \le \min\left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{25}$$

$$|\check{\mu}_i(t) - \mu_i| \le \left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{26}$$

$$|\hat{F}_i(t) - F_i| \le \left\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\right\}, \tag{27}$$

and:

$$\hat{r}_i(t) \ge r_i. \tag{28}$$

$$\check{\mu}_i(t) \le \mu_i. \tag{29}$$

$$\hat{F}_i(t) \ge F_i. \tag{30}$$

Then, if $\mathcal{O}_t$ holds, we have that:

$$\begin{aligned}\sum_{i\in\mathcal{P}_t} w_i\hat{r}_i(t) &\ge \alpha\cdot\max_{\mathcal{Q}'\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}'} w_i\hat{r}_i(t) \ge \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i\hat{r}_i(t)\\ &\ge \alpha\cdot\max_{\mathcal{Q}\subseteq\mathcal{A}_t}\sum_{i\in\mathcal{Q}} w_i r_i = \sum_{i\in\mathcal{P}_t} w_i r_i + \Delta(\mathcal{P}_t,\mathcal{A}_t),\end{aligned}$$

where $\mathcal{Q}' \subseteq \mathcal{A}_t$ is a subset satisfying $|\mathcal{Q}'| = L$, $\sum_{i\in\mathcal{Q}'}\check{\mu}_i(t) \le B$ and $\hat{F}_i(t) \ge F_{\min}, \forall i \in \mathcal{Q}'$, and $\mathcal{Q} \subseteq \mathcal{A}_t$ is a truly feasible subset satisfying $|\mathcal{Q}| = L$, $\sum_{i\in\mathcal{Q}}\mu_i \le B$ and $F_i \ge F_{\min}, \forall i \in \mathcal{Q}$. When $\mathcal{N}_t$ holds, using (29) and (30), we obtain $\mathcal{Q} \subseteq \mathcal{Q}'$ because:

$$\sum_{i\in\mathcal{Q}}\check{\mu}_i(t) \le \sum_{i\in\mathcal{Q}}\mu_i \le B,$$

$$\hat{F}_i(t) \ge F_i \ge F_{\min}, \forall i \in \mathcal{Q}.$$

Rearranging terms, we obtain:

$$\begin{aligned}\Delta(\mathcal{P}_t,\mathcal{A}_t) &\le \sum_{i\in\mathcal{P}_t} w_i\hat{r}_i(t) - \sum_{i\in\mathcal{P}_t} w_i r_i\\ &\le \sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i|.\end{aligned} \tag{31}$$

From (31) and the fact that $\Delta(\mathcal{P}_t,\mathcal{A}_t) \ge M$, we obtain: $\Delta(\mathcal{P}_t,\mathcal{A}_t) \le 2\sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i| - M$. This can be rewritten as:

$$\begin{aligned}\Delta(\mathcal{P}_t,\mathcal{A}_t) &\le 2\sum_{i\in\mathcal{P}_t} w_i|\hat{r}_i(t) - r_i| - M\\ &\le 2\sum_{i\in\mathcal{P}_t}\left(w_i|\hat{r}_i(t) - r_i| - \frac{M}{2|\mathcal{P}_t|}\right)\\ &= 2\sum_{i\in\mathcal{P}_t}\left(w_i|\hat{r}_i(t) - r_i| - \frac{M}{2L}\right),\end{aligned}$$

Then, using (25), the expression above becomes:

$$\begin{aligned}\Delta(\mathcal{P}_t,\mathcal{A}_t) &\le 2\sum_{i\in\mathcal{P}_t}\left(w_i\cdot\min\{2\sqrt{\frac{3\ln t}{2n_i(t-1)}}, 1\} - \frac{M}{2L}\right)\\ &= \sum_{i\in\mathcal{P}_t}\left(w_i\cdot\min\{\sqrt{\frac{24\ln t}{n_i(t-1)}}, 2\} - \frac{M}{L}\right).\end{aligned}$$

Finally, note that when $n_i(t-1) \le l_T(M)$, we have $w_i\cdot\min\{\sqrt{\frac{24\ln t}{n_i(t-1)}}, 2\} - \frac{M}{L} \le \kappa_T(M, n_i(t-1))$. When $n_i(t-1) > l_T(M)$, on the other hand, $w_i\cdot\min\{\sqrt{\frac{24\ln t}{n_i(t-1)}}, 2\} \le \sqrt{\frac{24\ln t}{n_i(t-1)}} \le \frac{M}{L}$, so $\min\{\sqrt{\frac{24\ln t}{n_i(t-1)}}, 2\} - \frac{M}{L} \le 0 = \kappa_T(M, n_i(t-1))$, which completes the proof. □

C. Proof of Theorem 2

We next present the complete proof of the constraint violation bound.

Theorem 2. For OQMPS, the total long-term violation $V(T)$ defined in (9) satisfies

$$V(T) \le 2\sqrt{24KLT\ln T} + 2\pi^2 KL + 2L\sqrt{T}.$$

Proof. We first decompose the total long-term violation into two parts:

$$V(T) = V_{cost}(T) + V_{fid}(T) \tag{32}$$

We define the long-term cost violation and long-term fidelity violation as follows:

$$V_{cost}(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T}\left(\sum_{i\in\mathcal{P}_t}\mu_i(t) - B\right)\right)^+\right], \tag{33}$$

$$V_{fid}(T) = \mathbb{E}\left[\left(\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}[F_{\min} - F_i(t)]\right)^+\right]. \tag{34}$$

At each time slot t, the oracle guarantees that the estimated cost and fidelity satisfy:

$$\begin{aligned}&\sum_{i\in\mathcal{P}_t}\check{\mu}_i(t) \le B,\\ &\hat{F}_i(t) \ge F_{\min}, \forall i \in \mathcal{P}_t.\end{aligned} \tag{35}$$

Using (35), we decompose the violations into two parts.

$$\begin{aligned}\sum_{i\in\mathcal{P}_t}\mu_i(t)-B&=\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\check{\mu}_i(t))+\underbrace{\sum_{i\in\mathcal{P}_t}\check{\mu}_i(t)-B}_{\leq 0},\\ \sum_{i\in\mathcal{P}_t}[F_{\min}-F_i(t)]&=\sum_{i\in\mathcal{P}_t}[(\hat{F}_i(t)-F_i(t))+\underbrace{(F_{\min}-\hat{F}_i(t))}_{\leq 0}],\end{aligned} \tag{36}$$

Therefore, we have:

$$\sum_{i\in\mathcal{P}_t}\mu_i(t)-B\leq\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\check{\mu}_i(t)), \tag{37}$$

$$\sum_{i\in\mathcal{P}_t}[F_{\min}-F_i(t)]\leq\sum_{i\in\mathcal{P}_t}(\hat{F}_i(t)-F_i(t)). \tag{38}$$

Recall that $\check{\mu}_i(t)=\bar{\mu}_i(t-1)-\mathrm{rad}_i(t)$ and $\hat{F}_i(t)=\bar{F}_i(t-1)+\mathrm{rad}_i(t)$. Thus the cumulative deviation is:

$$\begin{aligned}\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\check{\mu}_i(t))&=\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\mu_i)}_{S_T^{cost}}\\ &+\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(\mu_i-\bar{\mu}_i(t-1))}_{\mathcal{E}_T^{cost}}+\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)}_{\mathcal{R}_T}.\end{aligned} \tag{39}$$

and

$$\begin{aligned}\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(\hat{F}_i(t)-F_i(t))&=\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(F_i-F_i(t))}_{S_T^{fid}}\\ &+\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}(\bar{F}_i(t-1))-F_i)}_{\mathcal{E}_T^{fid}}+\underbrace{\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)}_{\mathcal{R}_T}.\end{aligned} \tag{40}$$

Hence:

$$\begin{aligned}V_{cost}(T)&\leq\mathbb{E}[(S_T^{cost})^+]+\mathbb{E}[(\mathcal{E}_T^{cost})^+]+\mathbb{E}[(\mathcal{R}_T)^+],\\ V_{fid}(T)&\leq\mathbb{E}[(S_T^{fid})^+]+\mathbb{E}[(\mathcal{E}_T^{fid})^+]+\mathbb{E}[(\mathcal{R}_T)^+].\end{aligned} \tag{41}$$

To bound the stochastic fluctuation terms $\mathbb{E}[(S_T^{cost})^+]$ and $\mathbb{E}[(S_T^{fid})^+]$, we apply Lemma 6 to $W_t^{cost}=\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\mu_i)$ and $W_t^{fid}=\sum_{i\in\mathcal{P}_t}(F_i-F_i(t))$. We verify that $\{W_t^{cost}\}$ and $\{W_t^{fid}\}$ are martingale difference sequences:

$$\begin{aligned}\mathbb{E}[W_t^{cost}|H_{t-1}]&=\mathbb{E}\left[\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\mu_i)\middle|H_{t-1}\right]\\ &=\sum_{i\in\mathcal{P}_t}\mathbb{E}[\mu_i(t)-\mu_i|H_{t-1}]=\sum_{i\in\mathcal{P}_t}(\mathbb{E}[\mu_i(t)]-\mu_i)=0.\end{aligned}$$

$$\begin{aligned}\mathbb{E}[W_t^{fid}|H_{t-1}]&=\mathbb{E}\left[\sum_{i\in\mathcal{P}_t}(F_i-F_i(t))\middle|H_{t-1}\right]\\ &=\sum_{i\in\mathcal{P}_t}\mathbb{E}[F_i-F_i(t)|H_{t-1}]=\sum_{i\in\mathcal{P}_t}(F_i-\mathbb{E}[F_i(t)])=0.\end{aligned}$$

and

$$|W_t^{cost}|=\left|\sum_{i\in\mathcal{P}_t}(\mu_i(t)-\mu_i)\right|\leq\sum_{i\in\mathcal{P}_t}|\mu_i(t)-\mu_i|\leq L.$$

$$|W_t^{fid}|=\left|\sum_{i\in\mathcal{P}_t}(F_i-F_i(t))\right|\leq\sum_{i\in\mathcal{P}_t}|F_i-F_i(t)|\leq L.$$

Therefore, we have:

$$\mathbb{E}[(S_T^{cost})^+]\leq L\sqrt{T}. \tag{42}$$

$$\mathbb{E}[(S_T^{fid})^+]\leq L\sqrt{T}. \tag{43}$$

To bound the estimation error terms $\mathbb{E}[(\mathcal{E}_T^{cost})^+]$ and $\mathbb{E}[(\mathcal{E}_T^{fid})^+]$, we decompose them into the event $\mathcal{N}_t$ and its complement:

$$\begin{aligned}\mathbb{E}[(\mathcal{E}_T^{cost})^+]\leq&\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right]\\ &+\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right]\end{aligned}$$

and

$$\begin{aligned}\mathbb{E}[(\mathcal{E}_T^{fid})^+]\leq&\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right]\\ &+\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right]\end{aligned}$$

Therefore, on the event $\mathcal{N}_t$, we have:

$$\begin{aligned}&\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\mathcal{N}_t)\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right]\leq\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right]\\ &=\mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\sqrt{\frac{3\ln t}{2n_i(t-1)}}\right]\\ &\leq\mathbb{E}\left[\sum_{i\in\mathcal{P}_t}\sum_{s=1}^{n_i(T)-1}\sqrt{\frac{3\ln T}{2s}}\right]\leq\mathbb{E}\left[\sum_{i\in\mathcal{P}_t}\sqrt{6n_i(T)\ln T}\right]\\ &\leq\mathbb{E}\left[\sum_{i=1}^{K}\sqrt{6n_i(T)\ln T}\right]\leq\mathbb{E}\left[\sqrt{6K\ln T}\cdot\sqrt{\sum_{i=1}^{K}n_i(T)}\right]\\ &=\sqrt{6KLT\ln T}\end{aligned} \tag{44}$$

On the complement $\neg\mathcal{N}_t$, we have:

$$\begin{aligned}\mathbb{E}\left[\sum_{t=1}^{T}\mathbb{I}(\neg\mathcal{N}_t)\cdot L\right]&\leq L\cdot\sum_{t=1}^{T}\mathbb{P}(\neg\mathcal{N}_t)\leq L\sum_{t=1}^{T}\frac{6K}{t^3}\\ &\leq L\sum_{t=1}^{T}\frac{6K}{t^2}\leq\pi^2KL.\end{aligned} \tag{45}$$

Combining (44) and (45), we have:

$$\mathbb{E}[(\mathcal{E}_T^{cost})^+]\leq\sqrt{6KLT\ln T}+\pi^2KL. \tag{46}$$

$$\mathbb{E}[(\mathcal{E}_T^{fid})^+]\leq\sqrt{6KLT\ln T}+\pi^2KL. \tag{47}$$

To bound the bonus term $\mathbb{E}[(\mathcal{R}_T)^+]$, we use the conclusion in (44):

$$\mathbb{E}[(\mathcal{R}_T)^+] = \mathbb{E}\left[\sum_{t=1}^{T}\sum_{i\in\mathcal{P}_t}\mathrm{rad}_i(t)\right] \le \sqrt{6KLT\ln T} \quad (48)$$

Therefore, the long-term cost and fidelity violations satisfy:

$$V_{cost}(T) \le \sqrt{24KLT\ln T} + \pi^2 KL + L\sqrt{T},$$

and

$$V_{fid}(T) \le \sqrt{24KLT\ln T} + \pi^2 KL + L\sqrt{T}.$$

Combining the established bounds for the cost violation and the fidelity violation, we immediately obtain:

$$\begin{aligned} V(T) &= V_{cost}(T) + V_{fid}(T) \\ &\le 2\sqrt{24KLT\ln T} + 2\pi^2 KL + 2L\sqrt{T}. \end{aligned}$$

□

Lemma 6. Let $\{H_t\}_{t=0}^T$ be a filtration, and $\{W_t\}_{t=1}^T$ a martingale difference sequence adapted to $\{H_t\}$, satisfying $\mathbb{E}[W_t \mid H_{t-1}] = 0$ for every time slot t. Suppose there exists a constant $C > 0$ such that $|W_t| \le C$ almost surely for every time slot t. Define the partial sum as $S_T = \sum_{t=1}^T W_t$. Then we have:

$$\mathbb{E}[(S_T)^+] \le C\sqrt{T}.$$

Proof. For any $\lambda > 0$, the standard Azuma-Hoeffding inequality for martingale differences with uniform bound C gives:

$$\mathbb{P}(S_T \ge \lambda) \le \exp\Big(-\frac{\lambda^2}{2TC^2}\Big).$$

For any real-valued random variable X, the expectation of its positive part can be rewritten as an integral over tail probabilities: $\mathbb{E}[X^+] = \int_0^\infty \mathbb{P}(X \ge \lambda)d\lambda$. Setting $X = S_T$ and plugging in the exponential upper bound, we obtain:

$$\mathbb{E}[(S_T)^+] \le \int_0^\infty \exp\Big(-\frac{\lambda^2}{2TC^2}\Big)d\lambda = \frac{1}{2}\sqrt{2\pi TC^2} \le C\sqrt{T}$$

□